

Cloud Database & API Recommendations (Free Tiers 2026)

1. Relational & Geospatial Database (PostgreSQL + PostGIS)

Your project needs PostGIS to store exact Lat/Long coordinates and perform spatial queries for your outbreak heatmaps.

Aiven (Best): <https://aiven.io/postgresql>

Offers an always-free managed PostgreSQL instance that includes the PostGIS extension out of the box. 1GB storage, 1GB RAM, 1 CPU without needing a credit card.

Supabase: <https://supabase.com/>

A popular Backend-as-a-Service. You get a full Postgres database where you can easily enable PostGIS.

Neon: <https://neon.tech/>

A serverless Postgres provider with a great free tier. Supports PostGIS and has a branching feature.

2. Vector Database (For Clinical AI RAG)

Your architecture uses Llama-3 that needs to read medical manuals. You need a Vector Database to store those text embeddings.

Qdrant (Best): <https://qdrant.tech/>

The king of free-tier vector databases. They give you a 1GB free cluster forever. Very fast and integrates perfectly with Hugging Face and FastAPI.

Pinecone: <https://www.pinecone.io/>

Fully serverless and requires zero setup or management. Perfect for rapid hackathon building.

3. Backend API Hosting (Python FastAPI)

To connect your mobile app and dashboard, you need to host your Python FastAPI server.

Koyeb (Best): <https://www.koyeb.com/>

Highly recommended. Generous free tier for running Docker containers and web services. Very fast to deploy from GitHub.

Render: <https://render.com/>

A classic choice for Python hosting, but spins down after 15 mins of inactivity (first request takes 50s).

Hackathon Strategy Recommendation

1. Aiven (for PostgreSQL + PostGIS)
2. Qdrant Cloud (for Vector DB)
3. Koyeb (for FastAPI)